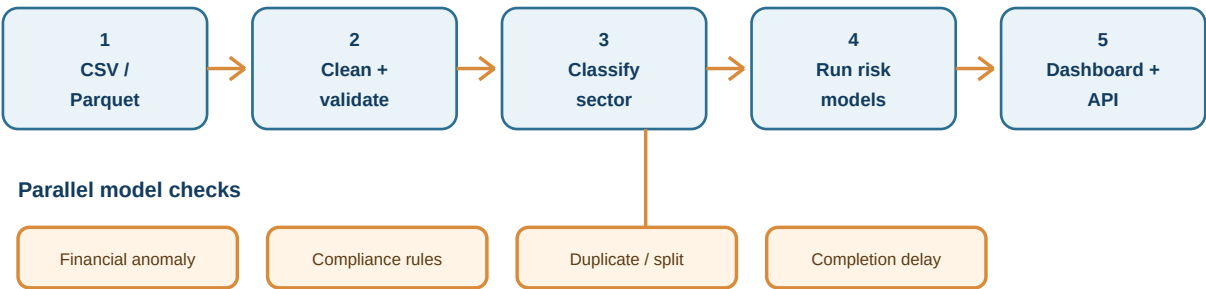


MPLADS Intelligence Platform

Complete website, data, workflow, model and operations guide

Purpose

This guide explains what the website does, how data moves through it, what every model checks, which conditions trigger findings, how risk is calculated, and what an operator should do with the results. It reflects the integrated 2023 MPLADS guidelines used by the current build.



Composite risk = compliance-first decision layer + explainable evidence

Reading this guide

The platform is an explainable screening and decision-support system. A model finding is evidence for review, not an automatic legal conclusion. Final sanction, rejection, inspection and release decisions remain with the authorized administration.

What the website does

This website is a decision-support and early-warning platform for monitoring MPLADS works. It joins allocation, recommendation, sanction, completion and expenditure records; calculates financial, compliance, completion and duplicate/split-work indicators; and presents the results for human review.

It does not legally adjudicate fraud, corruption or ineligibility. Its language is deliberately non-accusatory: risk indicator, potential anomaly, candidate duplicate, compliance gap and requires review.

End-to-end workflow

1. Raw government files are loaded from `data/raw`.
2. `src/preprocessing/cleaner.py` standardizes work IDs, text, dates, amounts and status fields.
3. `src/data/master_builder.py` joins the sanctioned works base to recommendations, completions and payment transactions.
4. Work descriptions are classified by `work_classifier.py` and `sector_classifier.py`.
5. Material context is calculated conservatively by `material_context.py` when a description contains explicit material quantities.
6. Financial, duplicate, compliance and completion-risk engines run over the master dataset.
7. `composite_risk_engine.py` merges model outputs, applies compliance-first priority, and writes `master_project_risk_scores.parquet`.
8. FastAPI serves the final data to the existing React/TypeScript UI.
9. Officials inspect the queue, open a work detail record, read the evidence explanation, and decide what manual verification is required.

Data foundation

Dataset	Purpose
T1 Allocated limits	MP/constituency entitlement and allocation context
T2 Calamity consents	Calamity-related consent amounts and dates
T3 Works recommended	Recommendation date, proposed description and amount
T4 Works sanctioned	Main work-level sanctioned record and sanction amount
T5 Works completed	Completion date, completed disbursement and evidence-image flag
T6 Expenditure	Payment date, amount and payment status

The pipeline creates processed Parquet files, an analytical master, model feature files and a final composite-risk master. Work IDs are normalized with regex extraction and deterministic fallback IDs when the source identifier is missing.

Website components

Overview / Executive Dashboard

Shows portfolio totals, sanctioned and disbursed amounts, completed works, risk distribution, state/category aggregation and review volume. It is for situational awareness and prioritization, not a final audit conclusion.

Risk Intelligence Monitor

Provides a paginated review queue. Records can be filtered by state, constituency, category, severity, search text and component risk thresholds. Sorting can focus on composite, financial, duplicate, compliance or schedule risk.

Project Detail / 360-degree view

Shows the work identity, description, sanction and expenditure values, completion/evidence information, component scores, explanations, peer baseline, schedule indicators and candidate duplicates.

Financial Analytics

Shows expenditure outliers, peer comparison, historical median context, anomaly type, deviation, sample size, unit-price context when available and a plain-language explanation of why a record requires review.

Compliance Monitor

Shows guideline and data-integrity conditions, triggered rules, severity and the required follow-up. Compliance is the highest-priority component.

Duplicate Inspector

Shows candidate pairs side by side, including descriptions, sector/category compatibility, dates, IDs, similarity, timing and split-work investigation evidence.

Schedule & Progress Risk

Compares elapsed or completed duration and expenditure progress against historical sector/category completion behavior and the one-year general guideline target.

MP Intelligence

Provides portfolio-level work counts, expenditure totals, completion counts and suspicious-work review records for the selected existing portfolio filters. It does not introduce national/state/constituency selector UI from the source application; it preserves the target application's existing interface.

Data Sync & System Status

Reports snapshot history, changes, last synchronization, next scheduled synchronization and operational status.

Model Monitoring

Reports the registered financial model, version, run history and tracked anomaly metrics from the local MLflow-style registry metadata.

AI Assistant

Remains the existing target-application assistant and is not replaced by the source project's UI.

Model and rule reference

1. Work classification and sector-cost model

Code: `src/modules/work_classifier.py`, `src/modules/sector_classifier.py`

Purpose: Convert free-text work descriptions into consistent groups and MPLADS sectors so that comparisons use relevant peers instead of unrelated works.

Inputs: description/sanctioned work description, source work category, state, sanction amount and explicit quantity/unit text.

Detection logic:

- Normalize text, punctuation and common variants.
- Apply high-confidence phrase rules for roads, bridges, water, lighting, education, healthcare, drainage, irrigation, buildings, sanitation, sports and other groups.
- If enough high-confidence examples exist, use TF-IDF features and a supervised text classifier for uncertain descriptions.
- Map recognized subcategories to the sector reference matrix.
- Extract quantities such as pieces, units, kilometres, metres, square feet and similar explicit units without inventing quantities.
- Apply the regional multiplier and reference cost data only as contextual information.

Outputs: `sector_id`, `main_sector`, `ai_work_category`, `effective_work_category`, confidence, matched keywords, classification reason, quantity, unit, reference unit/total cost and cost range fields.

2. Material context model

Code: `src/modules/material_context.py`

Purpose: Add a low-priority material-cost context signal where the description explicitly names a material and gives a material-native quantity.

Inputs: description, state, sanction/recommendation year and material benchmark reference data.

Detection logic:

- Detect cement, TMT steel, aggregate and brick indicators.
- Detect explicit quantities such as kg, tonne, bags or pieces.
- Match state/material/year benchmarks when available.
- Calculate a reference material cost only when a reliable quantity and price exist.

Outputs: detected materials, source of detection, quantity, benchmark price, reference material cost, context score, availability status and explanation.

Material context contributes only a small supporting portion of financial risk. It never replaces the completed-work financial baseline.

3. Financial anomaly model

Code: `src/modules/financial_anomaly.py`

Purpose: Identify unusual expenditure or cost patterns against comparable completed works.

Inputs: sanctioned amount, effective expenditure, completion status/date, constituency, sector/category, description quantity/unit, payment count, dates, reference-cost context and data-integrity fields.

Historical baseline: Completed works with positive expenditure are the primary historical reference. The model groups by classified category/sector and constituency. It calculates median, mean, quartiles, IQR, sample size and upper fence.

Unit-price branch: If the description contains a reliable quantity and unit, the model calculates `current_expenditure / quantity` and compares it with the historical median unit price for comparable completed works.

No-quantity branch: If there is no reliable quantity, it compares total current expenditure with the median total expenditure of comparable completed works. It does not create an artificial unit price.

ML branch: Isolation Forest uses normalized financial features including unit/total cost context, expenditure, peer ratio, payment count, expenditure-to-sanction ratio and quantity context. This is a supporting signal, not the sole decision.

Risk conditions: high peer deviation, high percentile, IQR upper-fence breach, expenditure above sanction, invalid financial values, date-order errors, missing core information, sector cost bounds and unusual ML pattern.

Outputs: anomaly flag/type/severity, financial score/level, expected median, current expenditure, deviation percentage, sample size, comparison basis, sector/category, constituency, comparable count, unit, quantity, current unit price, historical median unit price, confidence, explanation, risk description and structured reasons.

4. Compliance and guideline model

Code: `src/modules/compliance_engine.py`

Purpose: Apply deterministic, explainable administrative, financial-integrity and 2023 MPLADS guideline checks.

Inputs: recommendation/sanction/start/completion dates, work status, description, constituency, sanctioned amount, expenditure, sector, work type and available work metadata.

Current conditions include:

- Sanction or rejection not issued within 45 days.
- Date ordering violations.
- Similar recommendation timing signal.
- Completion that is implausibly fast.
- Incomplete work beyond the general one-year period.
- Severe ongoing delay beyond the extended threshold.
- Status/completion-date mismatch.
- Expenditure exceeding sanction.

- Negative or invalid amount combinations.
- Missing dates, short description or missing constituency.
- Sector cost and completion-duration reference bounds.
- Normal minimum work amount below ₹2.5 lakh, marked for exception review rather than automatic ineligibility.
- Residential buildings, commercial/private works, operation/maintenance, grants/loans, relief-fund contributions, land acquisition, reimbursement, individual/family benefit, CSR pooling, religious works, Swagat Dwar/welcome gates and unauthorized-colony works.
- Repair/renovation review for the 10% annual authorization cap and reasonable gap requirement.
- Movable-asset review for eligible government/government-aided institution and committee conditions.

Each rule produces a rule ID, name, source, condition, threshold, required fields, pass/fail status, severity and explanation. Missing source fields are treated as evidence gaps requiring manual verification, not proof of wrongdoing.

5. Completion and schedule-risk model

Code: src/modules/delay_risk.py

Purpose: Identify works whose elapsed/completed duration or financial progress is materially worse than comparable historical behavior.

Inputs: sanction date, completion date, status, effective expenditure, sanction amount, classified sector/category, description and reference date.

Detection logic:

- Calculate completed duration for historical completed works.
- Calculate sector/category median completion period and sample size.
- Use that historical median as expected completion period, falling back to the general one-year target when needed.
- Compare current elapsed/completed period, expenditure progress and expected timeline progress.
- Add overdue, progress-gap, peer-progress, one-year and calamity-duration signals.

Outputs: expected completion period, actual/elapsed period, sector median completion period, deviation, schedule/completion score and category, overdue days, progress gap and explanation.

6. Duplicate and possible split-work model

Code: src/modules/duplicate_detection.py

Purpose: Identify candidate duplicates or possible split works for manual verification.

Inputs: sanctioned description, normalized keywords, state, constituency, sector/category, work ID, sanction date and amount.

Detection logic:

- Compare works within the same state/constituency/category block.
- Normalize descriptions and calculate TF-IDF/cosine similarity.
- Use the same sanctioned description as the primary trigger for stronger investigation.
- Check same/compatible sector, timing within 180 days, numeric work-ID distance of up to 10 and surrounding record distance of up to 5 where available.
- Avoid labeling every similar description as a duplicate.
- Mark the pattern as suspected duplicate/possible split work and require manual verification.

Outputs: pair IDs, descriptions, similarity score, categories/sectors, amounts, dates, same-description flag, days between works, ID/record proximity, split-work investigation flag, risk level and explanation.

7. Final composite-risk model

Code: src/risk/composite_risk_engine.py

The final score combines compliance, financial, duplicate and schedule signals with compliance-first weights. Compliance is the controlling priority. A critical compliance result cannot be diluted by lower financial, duplicate or schedule scores. Financial

risk is next, followed by duplicate/split-work risk and schedule context.

Final outputs: overall score/category, separate component scores/categories, highest-priority reason, risk description, recommended action, evidence summary and confidence.

The final wording always identifies the most important review reason first and preserves the individual component signals.

API reference

Endpoint	Purpose
/api/health	Service health and loaded-record count
/api/overview	Portfolio totals, risk distribution, state/category summaries
/api/mp-intelligence	Existing portfolio intelligence and suspicious works
/api/risk-monitor	Paginated, filterable review queue
/api/work-detail	Complete work-level detail and candidate duplicates
/api/duplicate-candidates	Pairwise duplicate/split-work candidates
/api/schedule-risk	Schedule and completion-risk analytics
/api/filters	State, constituency, MP, category and severity values
/api/analytics/material-context	Material context for one work
/api/analytics/classification	Work classification for one work
/api/analytics/sector-cost	Sector and reference-cost context for one work
/api/sectors	Sector reference matrix
/api/compliance/rules	Current 2023 guideline thresholds and prohibited categories
/api/sync/status	Latest sync/snapshot status
/api/sync/history	Historical synchronization records
/api/model/status	Model registry and production status
/api/model/experiments	Model run summaries

Operating the project

Install dependencies with `pip install -r requirements.txt`. Run the complete pipeline with `python src/run_pipeline.py`. Start the backend with `python src/backend/app.py` or Uvicorn. Start the frontend from frontend using its package scripts.

The regenerated current dataset contains 79,068 sanctioned works. Always treat the model outputs as prioritization evidence. A reviewer should validate source documents, approvals, site conditions, user-agency commitments, procurement records, payment vouchers and applicable exceptions before taking administrative action.

Important limitations

- Text rules cannot prove ownership, location, agency eligibility or statutory clearance when those fields are absent.
- Sector cost ranges are reference context, not legal ceilings unless the applicable guideline or sanctioned schedule of rates makes them so.
- A missing photo or field is an evidence gap, not evidence that a work is fraudulent.
- Duplicate models identify patterns for investigation; they do not establish fraud or intentional splitting.
- The supplied document is named `mplads_2023_guidelines_including_changes.pdf`; its clauses and amendments are the compliance reference used by the current implementation.